Covariance and Correlation

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Descriptive statistics; Variance.
- **Leads to:** Feature selection; Correlation significance test; PCA; Ordinary least squares (closed form); Multivariate normal distribution.

1. Overview

Key point: Covariance gives the direction of the linear (straight-line) relationship between two numerical features; correlation divides it by the two standard deviations, which adds the strength and removes the effect of units.

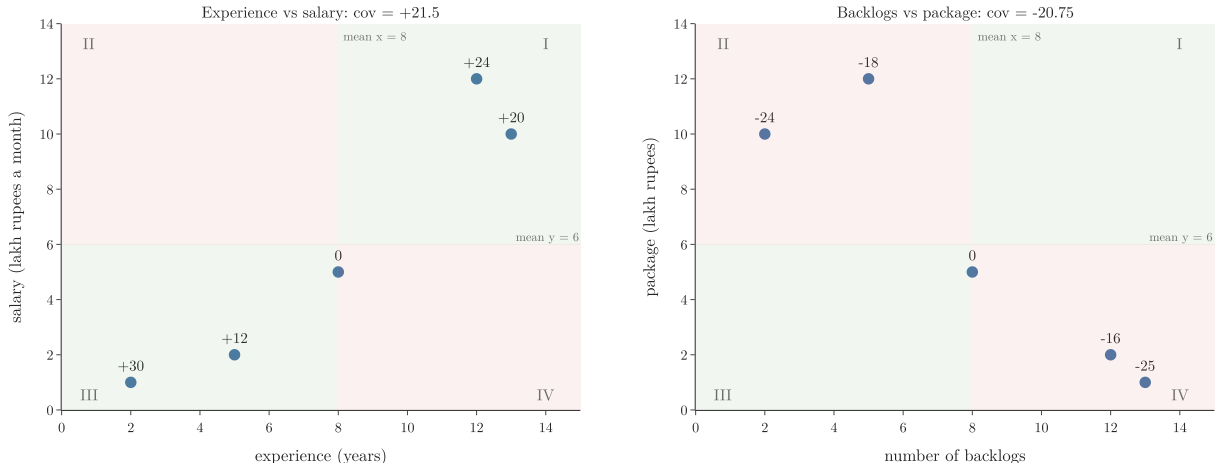


Figure 1: Covariance by quadrants: each point's product of distances from the two means

A **feature** (G-772) is one variable of the data, one column of the table; an **observation** (G-1374) is one record, one row. A **scatter plot** (G-1749; one point per observation, with one feature along each axis) shows by eye whether two numerical features rise together. Covariance and correlation turn that picture into a number. Figure 1 shows the idea behind both. Each dot is one observation. The two mean lines (a vertical line at the mean of the first feature and a horizontal line at the mean of the second) cut the plot into four quadrants, and each point votes positive or negative depending on its quadrant.

2. From mean to variance to covariance

> **Key point:** The mean gives the centre, variance gives the spread of one feature, and covariance gives how two features move together.

Each measure fixes a blind spot of the one before:

- **Mean** (G-1203): gives the centre of a feature, but $-10, 0, 10$ and $-20, 0, 20$ have the same mean, 0.
- **Variance** (G-2074): gives the spread, and tells those two apart (see the measures of dispersion). But it looks at one feature at a time.
- **Covariance** (G-496): the points $(-1,-1), (0,0), (1,1)$ rise from left to right, and $(-1,1), (0,0), (1,-1)$ fall. Their x and y variances are identical (2/3 each), as the PCA step by step (section 3.1) shows. Only a measure that uses both features together can tell them apart (Figure 2).

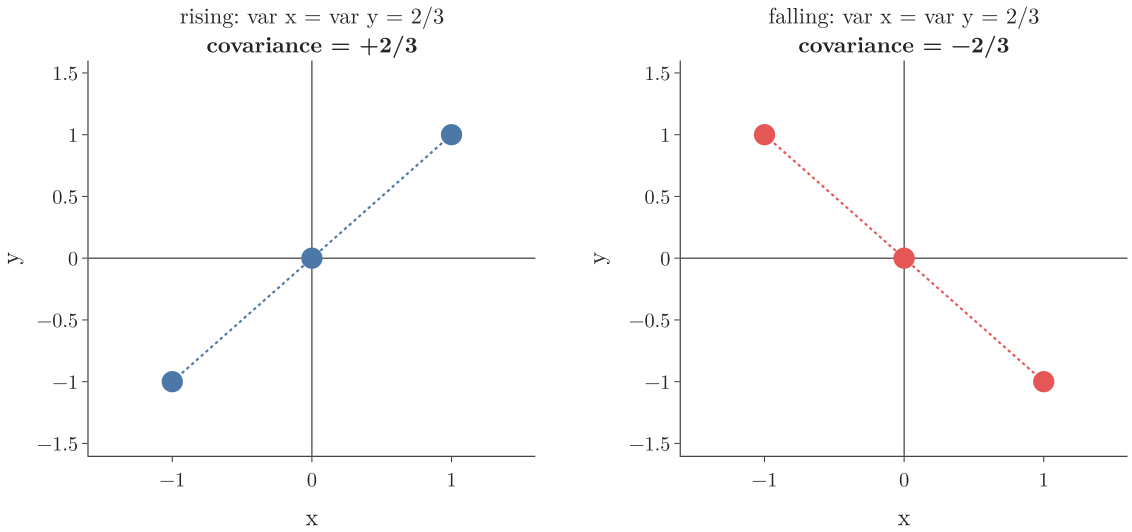


Figure 2: Two tiny datasets with the same means and the same variances, 2/3 for each feature. The rising one has covariance +2/3, the falling one −2/3.

Figure 2 can be checked line by line. In both datasets the means are 0, so each distance from the mean is just the value itself. The variance is the average squared distance, and the covariance is the average product of the two distances (both divide by $n = 3$ here).

Point	x	y (rising)	$x \times x$	$y \times y$	$x \times y$ (rising)	y (falling)	$x \times y$ (falling)
1	-1	-1	1	1	1	1	-1
2	0	0	0	0	0	0	0
3	1	1	1	1	1	-1	-1

Point	x	y (rising)	$x \times x$	$y \times y$	$x \times y$ (rising)	y (falling)	$x \times y$ (falling)
Sum			2	2	**2**		-2

Variance of x: 2/3. Variance of y: 2/3, in both datasets. Covariance, rising: 2/3. Covariance, falling: $-2/3$.

3. Covariance

> **Key point:** Covariance averages the product of each point's distances from the two means; positive means the features rise together, negative means one falls as the other rises, near zero means no straight-line relationship.

Do two features rise together? Take five employees (a sample) with their years of experience x and monthly salary y in lakh rupees. Salary rises with experience, but how do we turn that into one number? The idea: measure how far each employee sits from the average employee on both features, multiply the two distances, and average the products. This is **covariance** (G-496). The PCA step by step (section 3.2) teaches it, and the bivariate analysis shows how its sign gives the direction of a linear relationship.

The data: experience $x = 2, 5, 8, 12, 13$ years and salary $y = 1, 2, 5, 12, 10$ lakh rupees. First the two means, one per line.

$$\bar{x} = \frac{2 + 5 + 8 + 12 + 13}{5} = \frac{40}{5} = 8 \text{ years}$$

$$\bar{y} = \frac{1 + 2 + 5 + 12 + 10}{5} = \frac{30}{5} = 6 \text{ lakh rupees}$$

The bar over a letter means "mean of that feature". Then, for each employee, the distance from each mean and the product of the two distances:

Employee	x	y	$x - \bar{x}$	$y - \bar{y}$	Product
1	2	1	$2 - 8 = -6$	$1 - 6 = -5$	$(-6)(-5) = 30$
2	5	2	$5 - 8 = -3$	$2 - 6 = -4$	$(-3)(-4) = 12$
3	8	5	$8 - 8 = 0$	$5 - 6 = -1$	$(0)(-1) = 0$
4	12	12	$12 - 8 = 4$	$12 - 6 = 6$	$(4)(6) = 24$
5	13	10	$13 - 8 = 5$	$10 - 6 = 4$	$(5)(4) = 20$
Sum					$30 + 12 + 0 + 24 + 20 = 86$

Add the products and divide by $n - 1 = 4$ (the reason is the same as for the sample variance, see the measures of dispersion, section 6):

$$\frac{86}{5-1} = 21.5$$

Figure 3 draws the same computation. Each employee's product is the area of a rectangle from the two mean lines to that employee's point. Watch the running sum grow as each rectangle is added.

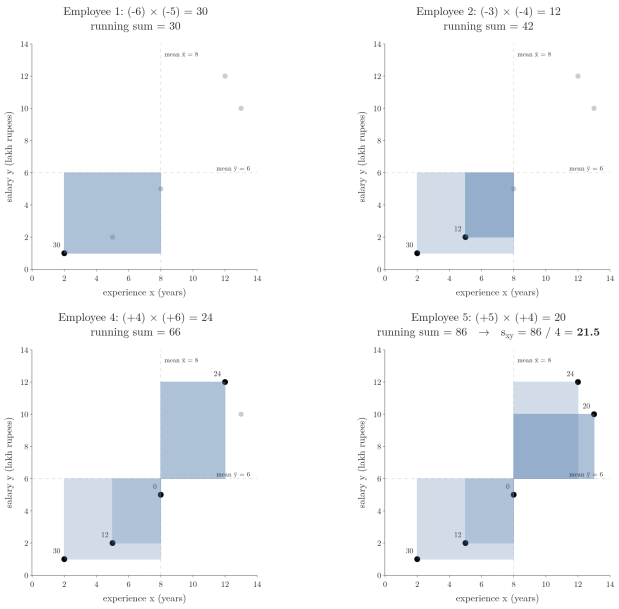


Figure 3: The five employees added one at a time. Each rectangle runs from the mean lines to the point; its area is the product in the table. The areas add up to 86, and 86 / 4 = 21.5.

- Employees 1 and 2 lie below and left of both means: two negative distances, a positive product (30 and 12).
- Employee 3 lies on the mean of x, so the rectangle has zero width and adds 0.
- Employees 4 and 5 lie above and right of both means: two positive distances, a positive product (24 and 20).

Every rectangle is positive, so the covariance is positive: more experience goes with a higher salary.

Now the formal version, with every symbol named. x_i is the experience of employee i (so $x_4 = 12$), y_i the salary ($y_4 = 12$), and $\sum_{i=1}^{n}$ means "add the terms for employee 1, 2, …, n". A **sample** (G-1731) of n employees divides by $n-1$. A whole **population** (G-1525) of N employees divides by N, and its means are written μ_x and μ_y ("mu").

Sample:

$$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Population:

$$\sigma_{xy} = \frac{1}{N}\sum_{i=1}^{N}(x_i - \mu_x)(y_i - \mu_y)$$

Check on the table: $n = 5$ and the sum is 86.

$$s_{xy} = \frac{86}{4} = 21.5$$

3.1 Reading covariance from the quadrants

> **Key point:** Points in quadrants I and III add to the covariance, points in II and IV subtract from it; whichever side holds more weight sets the sign.

The two mean lines split the scatter plot into four quadrants (Figure 1, left):

- **Quadrant I** (top right): both distances are positive, so the product is positive.
- **Quadrant III** (bottom left): both distances are negative, and negative times negative is positive.
- **Quadrants II and IV:** one distance is positive and one negative, so the product is negative.

So if most points lie in I and III, the covariance is positive; if most lie in II and IV, it is negative. If the points spread evenly over all four, the products cancel and the covariance is near 0. We can often guess the sign just by imagining the two mean lines on a scatter plot.

The right side of Figure 1 shows the opposite case. Five students have 2, 5, 8, 12 and 13 backlogs and packages of 10, 12, 5, 2 and 1 lakh rupees: more backlogs, lower package. The means, one per line:

$$\bar{x} = \frac{2 + 5 + 8 + 12 + 13}{5} = \frac{40}{5} = 8$$

$$\bar{y} = \frac{10 + 12 + 5 + 2 + 1}{5} = \frac{30}{5} = 6$$

Student	x (backlogs)	y (package)	$x - 8$	$y - 6$	Product
1	2	10	-6	4	-24
2	5	12	-3	6	-18
3	8	5	0	-1	0
4	12	2	4	-4	-16
5	13	1	5	-5	-25
Sum					-83

$$s_{xy} = \frac{-83}{5-1} = -20.75$$

A third case: if every student got the same package of 10 lakh, whatever their backlogs, then $y - \bar{y} = 0$ for every point. Every product is 0, and the covariance is exactly 0: no linear relationship.

A fourth case is a common interview trap: a covariance of 0 does not mean "no relationship". Take $x = -2, -1, 0, 1, 2$ and $y = x^2$, which is $4, 1, 0, 1, 4$. The means are 0 and 2.

Point	x	$y = x^2$	$x - 0$	$y - 2$	Product
1	-2	4	-2	2	-4
2	-1	1	-1	-1	1
3	0	0	0	-2	0
4	1	1	1	-1	-1
5	2	4	2	2	4
Sum					0

In Figure 4, each positive rectangle on one side is cancelled by an equal negative one on the other, so the sum is 0:

$$s_{xy} = \frac{0}{4} = 0$$

Yet y is completely fixed by x. Covariance sees only a linear trend: zero covariance means no linear relationship, and a curved one can still be there.

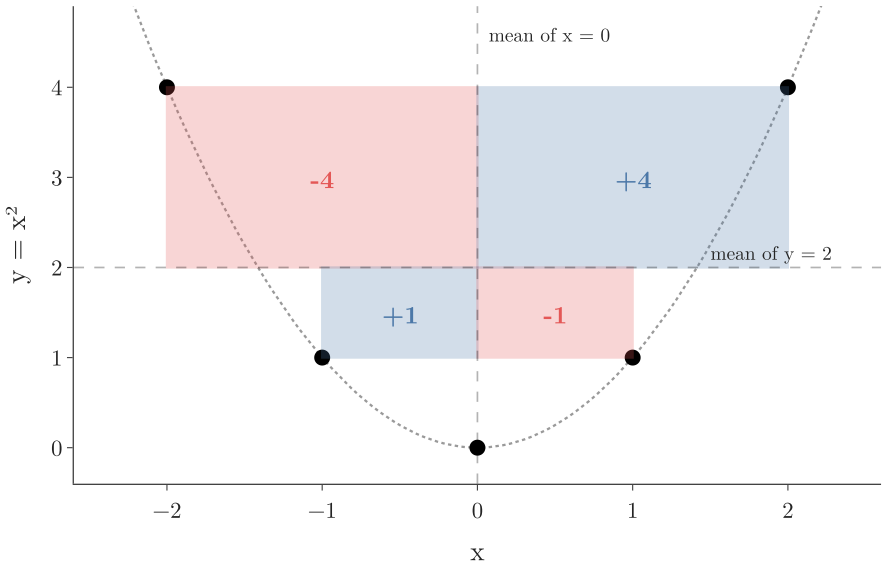


Figure 4: Five points on the curve $y = x^2$. The red rectangles (negative products) cancel the blue ones (positive products), so the covariance is 0 although the points follow the curve exactly

3.2 The flaw: covariance depends on the scale

> **Key point:** Multiplying a feature by a number multiplies the covariance by it too, so the size of a covariance says nothing about how strong the relationship is.

Covariance tells the direction of the **linear relationship** (G-1095), but not its **strength**: how closely the points follow a straight line. Its size depends on the units of the features.

Take the employees again and measure experience in months instead of years. Every x is multiplied by 12, so every distance from the mean is multiplied by 12, and so is every product:

Employee	Product in years	Product in months
1	30	$12 \times 30 = 360$
2	12	$12 \times 12 = 144$
3	0	0
4	24	$12 \times 24 = 288$
5	20	$12 \times 20 = 240$
Sum	86	1032

$$\text{cov} = \frac{1032}{4} = 258 = 12 \times 21.5$$

Measuring salary in rupees instead of lakhs multiplies the covariance by 100,000 as well: 2,150,000. The employees are the same; only the units changed.

The formal version: multiply every x by a number a and every y by a number c (here $a = 12$, $c = 1$), and every product, hence the covariance, is multiplied by $a \times c$.

$$\operatorname{cov}(a\,x,\ c\,y) = a\,c\,\operatorname{cov}(x, y)$$

Check with $a = 12$, $c = 1$ and $\operatorname{cov}(x, y) = 21.5$:

$$12 \times 1 \times 21.5 = 258$$

Figure 5 shows the problem with 40 random points; the three covariances are computed by code with the same formula as above, one product per point. The left panel plots a feature against itself, a perfect straight line, with covariance 1205. The middle panel plots x against a noisier y, a weaker relationship, with covariance 939. The right panel doubles both features: the picture is identical to the middle one, yet the covariance jumps to 3757, larger than for the perfect line.

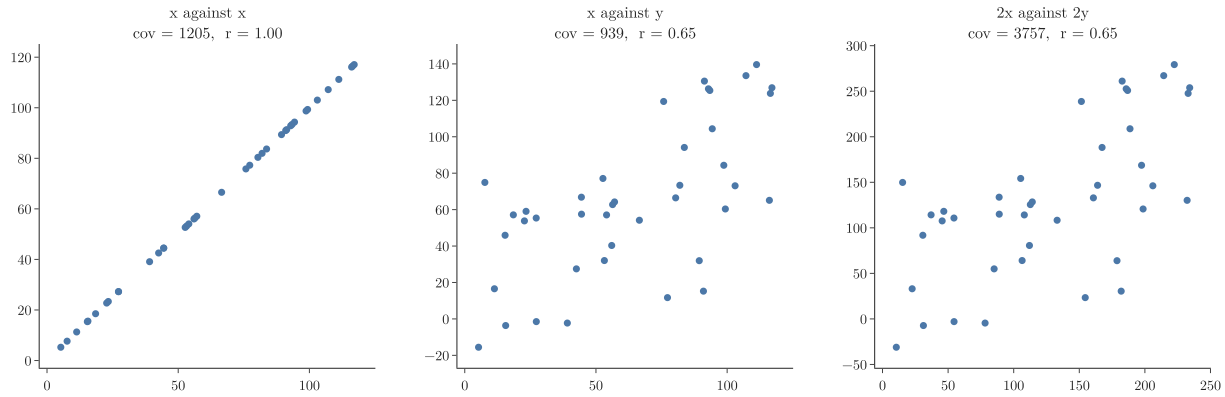


Figure 5: Covariance changes with the scale; correlation does not

So a large covariance does not mean a strong relationship. Covariance is reliable only for its sign. Its main use is as the building block of correlation.

> **Extra:** The covariance of a feature with itself is its variance. With $y = x$ the formula becomes the sample variance:
>
> $$\frac{\sum(x_i - \bar{x})(x_i - \bar{x})}{n - 1} = \frac{\sum(x_i - \bar{x})^2}{n - 1}$$
>
> The same fact explains why the covariance matrix of the PCA step by step has the variances on its diagonal, and why Figure 5's left panel shows the variance of x, 1205.

4. Correlation

> **Key point:** Correlation is covariance divided by the two standard deviations; it always lies between -1 and +1, gives both direction and strength, and does not change with the units.

The covariance 21.5 mixes years and lakh rupees, so its size means nothing by itself. To remove the units, divide it by how spread out each feature is: the **standard deviation** (G-1871) of x and of y. The result is the **correlation** (G-490).

The standard deviation is the square root of the average squared distance from the mean (with $n-1$ for a sample). Using the distances from the table in section 3, one step per line:

$$s_x^2 = \frac{(-6)^2 + (-3)^2 + 0^2 + 4^2 + 5^2}{4}$$

$$s_x^2 = \frac{36 + 9 + 0 + 16 + 25}{4}$$

$$s_x^2 = \frac{86}{4} = 21.5$$

$$s_x = \sqrt{21.5} = 4.637 \text{ years}$$

$$s_y^2 = \frac{(-5)^2 + (-4)^2 + (-1)^2 + 6^2 + 4^2}{4}$$

$$s_y^2 = \frac{25 + 16 + 1 + 36 + 16}{4}$$

$$s_y^2 = \frac{94}{4} = 23.5$$

$$s_y = \sqrt{23.5} = 4.848 \text{ lakh}$$

Then divide:

$$s_x \times s_y = 4.637 \times 4.848 = 22.48$$

$$r = \frac{21.5}{22.48} \approx 0.96$$

For the backlogs data, the same s_x and s_y apply (the backlogs are the same numbers, and the packages are the same numbers in another order), so

$$r = \frac{-20.75}{22.48} \approx -0.92$$

Both relationships are strong; one rises and one falls.

The formal version names the **Pearson correlation coefficient** r (G-1474), which appears in the understanding your data (section 9.1). With s_{xy} the covariance and s_x, s_y the two standard deviations,

$$r = \frac{s_{xy}}{s_x \, s_y}$$

For a population, the correlation is written ρ ("rho"):

$$\rho = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

The $n - 1$ in the covariance and in the two standard deviations cancel, so both versions give the same number. Check, as above:

$$\frac{21.5}{4.637 \times 4.848} \approx 0.96$$

4.1 Reading a correlation

> **Key point:** The sign gives the direction and the distance from 0 gives the strength; +1 and -1 are perfect straight lines.

The scale of r, from -1 to $+1$, is read as in the understanding your data (section 9.1): the sign gives the direction, and the closer $|r|$ is to 1, the closer the points lie to a straight line, so the stronger the relationship. Unlike covariance, which can be any number (1205, 3757, -5000), r never leaves that range.

Strength has a practical meaning: prediction. When the points lie close to a line, knowing x pins y down to a narrow range; when they scatter widely, the same x leaves a wide range of possible y values.

In Figure 5, x against itself gives $r = 1.00$. x against y gives $r = 0.65$: positive, but the points scatter around the line, so it is clearly below 1.

Figure 6 sweeps r from -1 to $+1$ on the same 60 random points. Watch the cloud:

- at $r = -1$ and $r = +1$ every point lies on one straight line;
- as $|r|$ falls to 0.9, 0.6 and 0.3, the points spread further from the line;
- at $r = 0$ the cloud has no tilt at all.

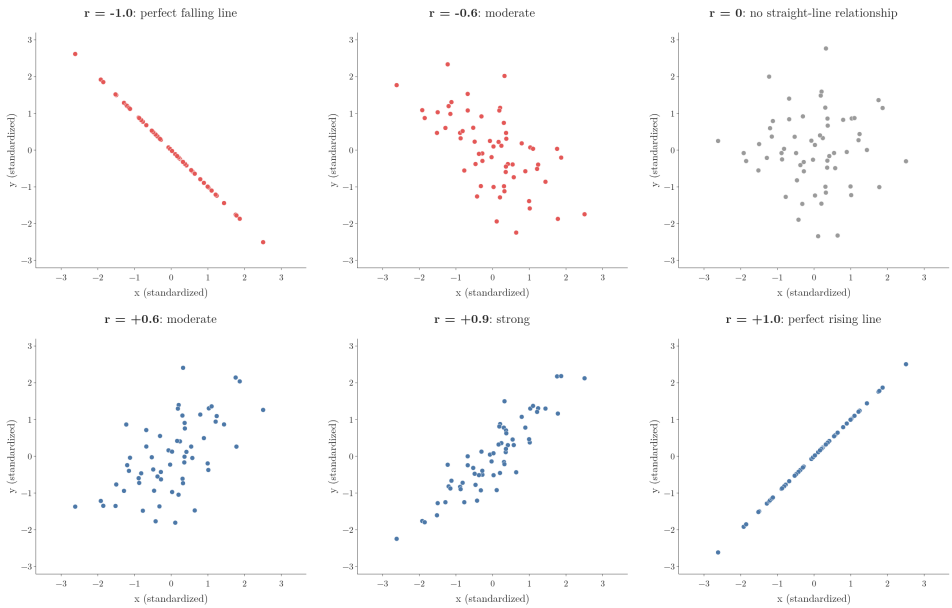


Figure 6: The same 60 random points with their correlation set to -1, -0.9, -0.6, -0.3, 0, 0.3, 0.6, 0.9 and $+1$ (standardized features). Red clouds fall, blue clouds rise, the grey cloud at 0 has no tilt.

> **Extra:** Rules of thumb put names on r. One common scale calls $|r|$ of 0.7 to 0.9 strong, 0.4 to 0.6 moderate and 0.1 to 0.3 weak (Akoglu 2018, Table 1). On this scale, the employees' $r = 0.96$ is strong and Figure 5's $r = 0.65$ is moderate.

4.2 Few points can fake a strong correlation

> **Key point:** A straight line passes through any two points, so two observations always give $r = +1$ or -1; a correlation is only as trustworthy as the amount of data behind it.

Take only the first two employees, $(2, 1)$ and $(5, 2)$. One step per line:

$$\bar{x} = \frac{2+5}{2} = 3.5$$

$$\bar{y} = \frac{1+2}{2} = 1.5$$

$$s_{xy} = \frac{(2-3.5)(1-1.5) + (5-3.5)(2-1.5)}{2-1}$$

$$s_{xy} = \frac{(-1.5)(-0.5) + (1.5)(0.5)}{1}$$

$$s_{xy} = \frac{0.75 + 0.75}{1} = 1.5$$

$$s_x = \sqrt{\frac{(-1.5)^2 + 1.5^2}{1}} = \sqrt{4.5} = 2.121$$

$$s_y = \sqrt{\frac{(-0.5)^2 + 0.5^2}{1}} = \sqrt{0.5} = 0.707$$

$$r = \frac{1.5}{2.121 \times 0.707} = \frac{1.5}{1.5} = 1$$

The result is $r = 1$, a "perfect" relationship, from two people. Any two points with different x and different y do the same, because one straight line always passes through both. With three or more points, landing on one line by chance becomes unlikely, and the more points there are, the more an observed r can be trusted.

So r answers "how strong is the relationship in this data?", not "how sure are we that it is real?". The second question needs the number of observations as well, and is answered by the correlation significance test in the choosing a hypothesis test.

4.3 Correlation does not depend on the scale

> **Key point:** Scaling a feature scales its standard deviation by the same amount, which cancels the change in the covariance.

In Figure 5, doubling both features quadrupled the covariance but left r at 0.65. Scaling by positive numbers never changes the correlation.

> **Extra:** Proof that correlation ignores the units.
> Example first: experience in months gives cov = 258 and a standard deviation 12 times larger:
>
> $$s_x = 12 \times 4.637 = 55.64$$
>
> $$r = \frac{258}{55.64 \times 4.848}$$
>
> $$r = \frac{258}{269.7} \approx 0.96$$
>
> as before. The 12 in the covariance and the 12 in s_x cancel. In general, multiplying x by a multiplies the covariance by a and the standard deviation of x by a too; the same holds for y with a number c. (Adding a constant changes nothing, since it moves the mean by the same amount.) For $a, c > 0$,

$$r(a\,x,\ c\,y) = \frac{a\,c\,\operatorname{cov}(x,y)}{(a\,s_x)(c\,s_y)}$$

$$r(a\,x,\ c\,y) = \frac{\operatorname{cov}(x,y)}{s_x\,s_y} = r(x,y)$$

If a or c is negative, the sign of r flips but its size stays.

Because it gives both the direction and the strength, and does not depend on units, correlation is the measure we use to study the linear relationship between two numerical features, for example before linear regression. Covariance is still needed, as the step that computes it.

Python: Covariance and correlation.

```
import numpy as np

x = np.array([2, 5, 8, 12, 13])
y = np.array([1, 2, 5, 12, 10])
np.cov(x, y)[0, 1]         # 21.5 (divides by n - 1)
np.corrcoef(x, y)[0, 1]    # 0.9565, about 0.96
```

Both return a 2 by 2 matrix; `[0, 1]` picks the value for the pair. In pandas, `df["a"].cov(df["b"])` and `df["a"].corr(df["b"])`. The Notebook (`MA-009-covariance-and-correlation.ipynb`) reruns the scale experiment of Figure 5.

5. Correlation does not imply causation

Key point: Two features can move together without one causing the other, often because a hidden third factor drives both.

Causation (G-359) is a cause-and-effect relationship: a change in one thing produces a change in the other. A correlation only says two things move together. The phrase "correlation does not imply causation" means that a correlation, however strong, is not evidence that one variable causes the other.

Two examples:

- **Firefighters and fire size.** At bigger fires, more firefighters are present: a strong positive correlation. Reading it as "more firefighters make bigger fires" is obviously wrong; the size of the fire decides how many firefighters are sent. Here the direction is clear, but in many datasets it is not.
- **Ice cream and drownings.** On days when more ice cream is sold, more people drown. Ice cream does not cause drowning. Hot weather drives both: when it is hotter, more people buy ice cream and more people go swimming (Figure 7). A third factor of this kind is what statisticians call a confounder (Freedman et al. 2007, ch. 2).

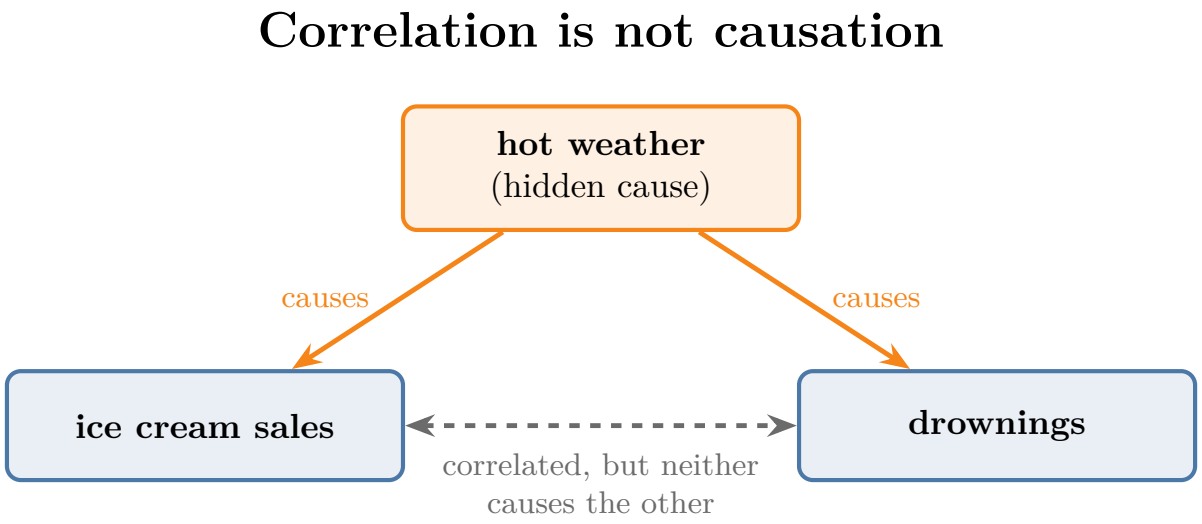


Figure 7: A hidden cause makes two features correlated

A hidden factor that drives two variables, like the weather here, is called a **confounding variable** (G-448), or confounder. Experience and salary are a subtler case: they are correlated, but the salary may come from skills that grow with experience, not from the years themselves.

Establishing causation needs more than data that happens to be collected: controlled experiments such as **randomised controlled trials** (G-1623), where a random half of the subjects gets a treatment and the other half does not, or carefully designed observational studies. Random assignment works because chance, not the subjects' own traits, decides who is treated, so hidden factors such as diet or weather even out between the two halves; a difference that remains can then be put down to the treatment (IMS §2.2.1). Until then, a correlation is a hint worth investigating, not a conclusion.

6. Summary

Measure	Formula	Range	Tells	Depends on units?	Why it matters
Covariance (sample)	$\sum(x_i - \bar{x})(y_i - \bar{y})/(n-1)$	any number	direction	yes	its size changes with the units, so read only its sign; it is the step that computes r
Correlation r	$\text{cov}(x,y)/(s_x s_y)$	-1 to $+1$	direction and strength	no	free of units, so it is the measure used to study a linear (straight-line relationship

Example	Covariance	r
Experience vs salary	21.5	0.96
Backlogs vs package	-20.75	-0.92
Same package for everyone	0	undefined (no spread in y)

- Points in quadrants I and III push the covariance up; II and IV push it down, because two distances with the same sign give a positive product. So the sign can be guessed by drawing the two mean lines on a scatter plot.
- $\text{cov}(ax, cy) = ac\,\text{cov}(x,y)$, but r stays the same, because the same factors scale the two standard deviations and cancel.
- The covariance of a feature with itself is its variance, which is why a covariance matrix has the variances on its diagonal.
- Zero covariance means no linear relationship; a curve such as $y = x^2$ can still be there, because covariance sees only a linear trend.
- Two points always give $r = \pm 1$, because one straight line passes through any two points; trust r only with enough data.
- Correlation does not imply causation; a confounding variable can drive both features, so a strong r is not evidence that one feature causes the other.
- Correlation turns the scatter plot into one number for direction and strength, so it is the measure to check, for example before linear regression.

7. Sources

Built from

- CampusX, "Session 39 - Descriptive Statistics Part 2 | DSMP 2023", YouTube, https://www.youtube.com/watch?v=1ndVC500-EU
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- StatQuest with Josh Starmer, "Pearson's Correlation, Clearly Explained!!!", YouTube, https://www.youtube.com/watch?v=xZ_z8KWkhXE

Other references

- Akoglu, H. (2018). User's guide to correlation coefficients. *Turkish Journal of Emergency Medicine*, 18(3), 91-93. Table 1.
- Çetinkaya-Rundel, M. and Hardin, J. (2021). *Introduction to Modern Statistics*. OpenIntro. §2.2.1, Principles of experimental design (randomization evens out confounding variables). https://openintro-ims.netlify.app/data-design
- Freedman, D., Pisani, R. and Purves, R. (2007). *Statistics*, 4th ed. W. W. Norton. Chapter 2, Observational Studies (association is not causation; confounding factors).

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Covariance (G-496)	A number that says how two numerical features move together: positive if they rise together, negative if one rises as the other falls. Its size depends on the units.
Correlation (G-490)	A number from -1 to +1 that says how strongly two numerical features rise or fall together along a straight line: the covariance divided by both standard deviations, so the units drop out.
Causation (G-359)	A cause-and-effect relationship: changing one thing changes the other.
Confounding variable (G-448)	A hidden factor that drives two variables and makes them correlated.
Randomised controlled trial (G-1623)	An experiment that assigns a treatment at random, to test causation.
Population covariance (σ_{xy}) (G-1523)	The covariance computed from every member of a population: the average product of the two columns' distances from their means, dividing by N; it shows whether the two columns move together.

Term	Meaning
Sample covariance (s_{xy}) (G-1724)	Covariance computed from a sample of n observations: the products of the two features' distances from their means, added up and divided by $n-1$; its sign says whether the two features rise together or move in opposite directions.
Strength of a relationship (G-1897)	How closely the points follow a straight line; measured by $\|r\|$.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Scatter plot (G-1749)	A chart with one dot per row and one numerical column on each axis; it shows by eye whether two features rise together, fall together or are unrelated.
Mean (G-1203)	The average of the values; the centre of the data.
Variance (of data) (G-2074)	The average squared distance of the values from their mean; the square of the standard deviation; divide by n for a population (NumPy default) or $n-1$ for a sample (pandas default).
Sample (G-1731)	The part of a population that we actually measure, such as 50,000 people asked about their salary instead of everyone in India; we study it because measuring the whole population is usually impossible, and use it to draw conclusions about the population.
Population (G-1525)	The entire group of individuals or objects we want to study.
Linear relationship (G-1095)	A relationship between two columns that follows a straight line.
Standard deviation (G-1871)	How far values typically lie from the mean; divide by $n-1$ for a sample (pandas default) or by n for a population (NumPy default).
Pearson correlation coefficient (Pearson's r) (G-1474)	The usual measure of correlation for linear (straight-line) relationships between two numerical columns, written r; the one `df.corr()` computes.